

Tim Umbach

Data Scientist

Experimentation, causal inference and product analytics



Hamburg, Germany

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Data scientist and applied econometrician specialising in experimentation and causal inference. I turn large-scale behavioural and transaction data into reliable evidence for product decisions, combining hands-on analysis in Python, SQL and Google Cloud with clear stakeholder guidance.

EXPERIENCE

Data Scientist, Experimentation & Product Analytics

Oct 2025 – present

bonprix Handelsgesellschaft mbH, Otto Group

Hamburg, Germany

- Design and analyse A/B tests end to end, from hypotheses and KPI definitions to power, randomisation checks, segmentation, effect estimation, and business interpretation.
- Analyse large-scale clickstream and transaction data in BigQuery and Python and build reusable workflows for experimentation and product analytics.
- Apply causal and robust statistical methods to constrained experiments and observational questions, including difference-in-differences, instrumental variables, panel models, resampling, and multiple-testing corrections.
- Advise product, design, marketing, and CRM teams on measurement strategy, uncertainty, and evidence-based prioritisation.
- Use Git and GitHub for reproducible development and Codex, Claude Code, and OpenCode for agentic engineering, including coding, review, documentation, and analytical tooling in Vertex AI.

IT Management Trainee, Digital Strategy

Nov 2024 – Jun 2025

Free and Hanseatic City of Hamburg, Senate Chancellery

Hamburg, Germany

- Coordinated cross-agency data and AI initiatives and contributed to governance for data procurement, classification, sharing, and reuse.
- Prepared strategic briefings and decision materials for senior leadership and supported interdepartmental workshops and committees.

Research Associate & Developer, Economics

Dec 2017 – Apr 2024

University of Cologne

Cologne, Germany

- Conducted empirical research in applied microeconomics, labour economics, crime, and illicit markets, using quasi-experimental designs, instrumental variables, panel data, and time-series methods.
- Built an automated data pipeline that collected darknet market data for approximately 18 months to study pricing in illegal drug markets.
- Taught microeconomics and quantitative methods, supervised student work, and developed adaptive STACK/Maxima exercises for large courses in economics, statistics, and mathematics.
- Supported research and teaching infrastructure through Python automation, web scraping, Linux administration, and reproducible data workflows.

International Development Service

Sep 2008 – Aug 2009

German Development Service / CSG

Hanoi, Vietnam

- Supported grant applications for a local NGO and provided English-language training to staff from several non-governmental organisations.

EDUCATION

Doctoral Studies in Economics

2017 – 2024

University of Cologne

Advanced coursework and research in empirical methods, causal inference, and machine learning; NLP summer school at Ruhr University Bochum.

Coursework average: 1.2

M.Sc. Quantitative Economics

2015 – 2017

Kiel University

Focus on statistics, econometrics, and public policy. Thesis: *Spatial Crime Patterns in Germany and their Socio-Economic Determinants*.

Final grade: 1.6

B.Sc. Economics

2009 – 2015

Humboldt University of Berlin

Focus on empirical methods and public policy, with art history as a minor. Thesis: *A Capital Allocation Puzzle within the EU?*

Final grade: 1.8

SKILLS

Experimentation & causal inference

A/B testing, experiment design, power and MDE analysis, randomisation diagnostics, difference-in-differences, instrumental variables, panel and time-series models, permutation and bootstrap methods, multiple testing

Programming & analytics

Python, SQL, BigQuery, R, Stata, Excel

Cloud & engineering

Google Cloud Platform, Vertex AI, Git, GitHub, Linux, web scraping, workflow automation

Stakeholder & delivery

Executive-ready insights, cross-functional consulting, requirements clarification, workshop facilitation, data storytelling, Jira, Kanban

Agentic engineering

Codex, Claude Code, OpenCode

Visualisation & reporting

Tableau, Power BI, LaTeX

SELECTED RESEARCH

- Hansen, P., Struth, L., Thon, M. and Umbach, T. (2021). *The Impact of the COVID-19 Pandemic on Teaching Outcomes in Higher Education*. ECONtribute Discussion Paper No. 073.
- Umbach, T. (2020). *A Vicious Cycle of Regional Unemployment and Crime? Evidence from German Counties*. Annual Conference of the German Economic Association.
- Fath, J., Hansen, P., Scheicher, C. and Umbach, T. (2019). *E-homework with Individual Feedback for Large Lectures*. 1st International STACK Conference.

ENGAGEMENT & LANGUAGES

2018 – 2019

2015 – 2017

Languages

Deputy member of the Faculty Council, University of Cologne

Founding board member, Pluralist Economics Kiel

German: native | English: C2, TOEFL iBT 113/120 | French: A2